

Implementation of Fully Integrated Continuous Antibody Processing: Effects on Productivity and COGm

Lindsay Arnold, Kenneth Lee, Joanna Rucker-Pezzini, and Jeong H. Lee*

The changing landscape of the biopharmaceutical market is driving a paradigm shift toward continuous manufacturing. To date, integrated continuous bioprocessing has not been realized as enabling technologies are nascent. In this work, a fully integrated continuous process is successfully demonstrated from pilot scale bioreactor to drug substance. Comparable product quality is observed between the continuous process and a 500 L fed-batch conventional process. The continuous process generated material at a rate of 1 kg of purified mAb every 4 days, achieving a 4.6-fold increase in productivity compared to the fed-batch process. A plant throughput analysis using BioSolve software shows that a fed-batch facility with $4 \times 12\,500$ L stainless steel bioreactors and purification train of the corresponding scale can be replaced by a continuous facility consisting of 5×2000 L single use bioreactors and smaller purification train, with a cost reduction of 15%.

1. Introduction

As the global biologics portfolio becomes more complex and biosimilars enter the market, one of the main challenges that the biopharmaceutical industry will face is the development of more cost effective, flexible manufacturing processes. In addition to biosimilar competition, the number of blockbuster biologics is expected to decline, which would further increase the pressure on cost of goods manufactured (COGm).^[1–3] Continuous processing has replaced batch manufacturing in other industries due to its improved efficiency, flexibility, and productivity,^[2,4–6] and now the biopharmaceutical industry is actively developing continuous manufacturing technologies to achieve these goals.^[7,8]

Continuous cell culture processes have been used for many years primarily for manufacturing labile proteins, such as enzymes or coagulation factors, where continuous harvest is required for optimal product quality.^[9–12] Maturation of perfusion-based continuous processes and recent advancements in continuous protein purification technologies^[13–16] have enabled development of fully-integrated continuous manufacturing. These processes have shown increased volumetric productivity while the economic analyses on COGm have

shown mixed results when compared to batch processing.^[17–19] Such technological advancements present opportunities to increase manufacturing flexibility and reduce capital and utility costs when building new manufacturing facilities.^[20,21] As demonstrated in this work, increased productivity of continuous processes allows for reduction of equipment size, which in turn allows for implementation of single-use technologies. In addition, further reduction of facility footprint can be achieved by elimination of hold vessels between unit operations. Continuous manufacturing also decreases costs associated with process failures. In a batch scenario, a failure in any single step may result in total loss of drug substance. If there is a process deviation in a continuous process, specifically with the bioreactor or

ATF unit, all material prior to the deviation would be delivered to the MCC, and if the process was recovered after the deviation, subsequent material can then be delivered. If there is a process deviation within the purification train, all material upstream of the failure point can be diverted to a temporary hold tank while the deviation is rectified. In both instances, the continuous process would be able to progress after process recovery. Therefore, the benefits of continuous processing extend beyond the parameters included in economic and throughput analyses.

In the evolving pursuit of continuous bioprocessing, many companies, such as Genzyme, Bayer, Janssen, Merck, Novartis, and Lonza, have put extensive effort into continuous process development, while some have incorporated continuous processing elements in the manufacture of their biologics.^[22–25] Academia has kept pace with the advancing needs of the industry as well (<https://www.boston.com/tag/business>, September 28, 2009). However, notable technology gaps in viral inactivation strategy and UFDF have limited the end-to-end processing capabilities. While many small-scale demonstrations have been done, there is limited published work on scaled up, multi-day operation, which begins to bridge the gap from concept to implementation.

This manuscript describes the fully integrated continuous manufacture of an IgG from production bioreactor through formulation unit operations using a combination of existing and novel technologies. A previously optimized perfusion process with continuous harvest was successfully integrated with a newly developed continuous purification process at pilot scale. Single-use technologies, such as the ATF, the bioreactor, filters, and membranes, were implemented to mimic a future single-use

Dr. L. Arnold, Dr. K. Lee, J. Rucker-Pezzini, Dr. J. H. Lee
Biopharmaceutical Development
MedImmune
1 MedImmune Way, 20878 Gaithersburg, MD, USA
E-mail: leej@medimmune.com

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facility. The steady-state cell culture methodology and the technologies used in continuous purification and formulation are discussed along with the process performance. COGm analysis is also presented to show the cost breakdown for hypothetical batch and continuous scenarios with various process durations. The model shows a potential COGm reduction of 15% with 2000 L continuous processing as compared to batch processing at the 12 500 L scale with 30-day run duration; however, the benefit of further increase in run duration on COGm diminishes significantly beyond 30 days.

In this proof of principle study, a fully integrated continuous process was successfully demonstrated.

2. Experimental Section

2.1. Cell Line and Inoculum Expansion

For this study, an in-house CHO derived cell line producing an IgG1 chimeric fusion protein was used. Cryopreserved cells were thawed in a proprietary animal protein free medium (APF) and split in shake flasks of increasing volume from 50 mL to 2 L in an incubator at 37 °C, 6% CO₂ and 120 rpm agitation with passive humidification until a sufficient volume of inoculum was obtained to inoculate the final seed vessel. Finally, 2 L shake flasks for inoculation of 2 L scale production bioreactors (Applikon, Dover, NJ, USA) or 15 L seed bioreactors (Bellco, Vineland, NJ, USA) for inoculation of 50 L scale production bioreactors (ThermoFisher Scientific, Waltham, MA, USA) were used.

2.2. Production in Perfusion Bioreactor

Cells were inoculated into either 2 L glass vessels or a 50 L single use bioreactor (SUB) for the production stage when a predetermined viable cell density (VCD) criterion was met in the final seed vessel. Two days after inoculation, perfusion was initiated at 1.5 volume perfused per reactor volume per day (VVD) with a proprietary APF perfusion medium. ATF 2 and disposable ATF 6 units (Repligen, Waltham, MA, USA) with the appropriately sized 0.2 µm polyethersulfone (PES) hollow fiber filters (Repligen) were used for the 2 L and 50 L-scale perfusion runs, respectively. The alternating tangential flow rate was set to ensure a minimum volumetric tangential flow to volumetric permeate flow ratio of 150:1 per vendor recommendation. When the VCD reached greater than 80×10^6 cells mL⁻¹, cell bleeding was initiated to maintain the VCD between 80 and 120×10^6 cells mL⁻¹. At this point permeate was collected for further processing. Bleed rates were calculated based on the calculated growth rate and current bleed rate and were adjusted once daily to target the projected VCD of 100×10^6 cells mL⁻¹ at 24 h after each sample time (Equation (1)). All other bioreactor set points were predetermined through a series of optimization studies in 2 L bioreactors with ATF 2.

Cell count was measured daily by ViCell (Beckman Coulter, Indianapolis, IN, USA). Gas and metabolite measurements were also made daily using RAPIDPoint 500 Blood Gas Systems (Siemens, Washington, D.C., USA).

Equation (1)

$$F_{\text{bleed}_n} = \left[1 - \frac{x_{\text{target}}}{x_n e^{\mu_n(t_{n+1}-t_n)}} \right] \frac{V_{\text{brx}}}{(t_{n+1} - t_n)}$$

Where

$$\mu_n = \frac{\text{LN} \left[\frac{x_n + \frac{x_n F_{\text{bleed}_{n-1}}(t_n - t_{n-1})}{V_{\text{brx}}}}{x_{n-1}} \right]}{t_n - t_{n-1}}$$

x_{target} : target viable cell density at steady state

x_n : viable cell density at $t = n$

F_{bleed_n} : bleed flow rates at $t = n$

μ_n : specific growth rate at $t = n$

t_n : sample time

V_{brx} : culture volume

F_{bleed_n} : bleed flow rates at $t = n$

A steady-state cell density within the range described above and productivity up to 90 days was demonstrated using this strategy with the same cell line in a previous experiment (data not shown). This knowledge was used in scaling up the process to 50-L.

2.3. Capture Multicolumn Chromatography

A Novasep BioSC Multicolumn Chromatography system (Novasep, Boothwyn, PA, USA) was used to perform continuous bind and elute chromatography. Five 2.2 cm Vantage L columns (Millipore, Burlington, MA, USA) were packed to a 5 cm bed height with MabSelect SuRe Protein A resin (GE Healthcare, Uppsala, Sweden). After loading to 29 grams of product per liter of resin, a column would be re-equilibrated, then washed with three column volumes of high salt buffer. Product was then eluted with low pH buffer with the peak collection criteria at 0.1 OD₂₈₀. The column was stripped with acetic acid, then washed with a caustic wash of hydroxide before regeneration. The product was eluted into a glass bottle containing multiple elution pools and was well mixed to ensure a homogenous solution for forward processing.

2.4. Viral Inactivation

Masterflex L/S Digital pumps (Cole-Parmer, Vernon Hills, IL, USA) were used for product, acid, and base addition. Well-mixed

Protein A eluate pool was diluted in-line with 1M acetic acid to reach the target pH range of 3.4–3.6. A static mixer (Koflo, Cary, IL, USA) was used to ensure mixing, and a pH meter (Mettler Toledo, Columbus, OH, USA) was used to monitor inline pH. The acidified mixture was continuously fed for the duration of the process through a 5 cm GE XK 50/30 column packed to a 20 cm bed height with Superdex 200 size exclusion resin from GE to provide the required residence time for viral inactivation. Following the column hold, the neutralization was performed with continuous addition of 1M Tris base into the 200 mL working volume break tank and mixed by a stir bar.

2.5. Filter Train

The filter train was comprised of a Millipore Millistak+[®] X0HC Pod Depth Filter, a Millipore Single Pass Tangential Flow Filtration (SPTFF) element containing Millipore Pellicon-3 30 kD membrane cassettes, a Millipore Express[®] SHC 0.5/0.2 μ m Hydrophilic Filter, a 10 mL Pall Mustang Q Anion Exchange Membrane (Pall Life Sciences, Westborough, MA, USA), and a 0.12 m² Planova[™] 20N (Asahi Kasei, Tokyo, Japan), all connected in series. Single use pressure sensors (Pendotech, West Windsor Township, NJ, USA) were used for pressure measurement.

2.6. Ultrafiltration and Diafiltration

Single Pass Diafiltration (SPDF) and SPTFF were performed according to the instructions given elsewhere.^[26] Four Pall Corporation Cadence Inline Concentrator 4-in-series membranes, with a molecular weight cutoff of 30 kD, were used. The module size was the smallest available at 0.065 m².

2.7. Analytics

Volumetric antibody titer was quantified by Protein A affinity chromatography, using a Protein A ImmunoDetection sensor cartridge (ThermoScientific, Ashville, NC, USA) coupled with an Agilent 1200 series HPLC (Agilent Technologies, Santa Clara, CA, USA). Peak areas relative to a reference standard calibration curve was used to calculate titers.

Drug intermediate sample protein concentration was measured using a NanoDrop[™] 2000 (Thermo Fisher Scientific, Wilmington, DE, USA) or by Solo VPE (C Technologies, Bridgewater, NJ, USA) at 280 nm.

Immunoassays were performed using a Gyros immunoassay platform (Gyros AB, Uppsala, Sweden). Aggregate was measured with HPSEC. Purity analysis was performed using a TSK-GEL G3000SWXL column (7.8 mm $\times$ 30 cm) (Tosoh Bioscience, King of Prussia, PA, USA) with an Agilent 1200 HPLC system.

The DNA determination assay was performed in 96-well plate format using a forward and reverse primer set specific for the Chinese hamster type 2 Alu equivalent sequence of the short interspersed repetitive element (SINE) family and Power SYBR[®] Green chemistry. A unique 64 base-pair product was generated from CHO genomic DNA calibrators and tested within 40 cycles

of amplification by PCR. DNA was extracted from test samples and a control using the DNA Extractor Kit[®] (Wako Chemicals USA, Richmond, VA, USA). Quantities of DNA in the samples above LLOQ were regressed from an assay specific standard curve generated by the software.

2.8. COGm and Throughput Modeling

COGm and throughput modeling of different manufacturing scenarios was executed in BioSolve software (version 7.5) (Biopharm Services Ltd, Chesham, UK). For fed-batch production, four 12 500 L stainless steel bioreactors operating in fed-batch mode and processed through a single purification train was used to predict cost per gram and throughput. These four bioreactors were staged to allow individual bioreactor harvests without requiring pooling. For continuous production, five 2000 L perfusion SUBs were simultaneously pooled and processed by one purification train. More details on the input parameters and the key assumptions are discussed in the results section.

3. Results

3.1. Setup of Fully Integrated Continuous Process

A fully integrated mAb production and purification process was constructed and experimentally operated as a pilot scale proof of principle. The original process design is summarized in Process Flow Diagram (PFD) in **Figure 1A**, and illustrates the batch monoclonal antibody platform. The integrated continuous process design is pictured in **Figure 1B**. Continuous upstream production was achieved with a 50 L perfusion SUB operated at 30 L. Perfusion medium exchange was initiated on day 2 of culture and a cell bleed was initiated, in accordance with Equation (1), once cells reached 80×10^6 cells mL⁻¹, as shown in **Figure 2A**. Steady state cell density ($80\text{--}100 \times 10^6$ cells mL⁻¹) was maintained until the end of the run (day 17 for 50 L SUB; > 30 days for 3 L bench scale). In contrast, fed-batch bioreactors using a bolus feeding strategy showed a peak viable cell density of 48×10^6 cells mL⁻¹. Fed-batch bioreactors were terminated after 14 days of culture. Viability for all bioreactor runs remained above 80%. The experiment at the 50 L scale was forcibly terminated ahead of schedule due to an unexpected failure in power. The NovaSep BioSC multicolumn chromatography system was used to enable continuous bind and elute chromatography for the Protein A capture step. NovaSep currently offers a bench scale unit that can simultaneously control six columns and operate at up to 30 mL min⁻¹ flow rates. However, based on break through curve optimization and the titer of the bioreactor, only a five column method was needed.

The Protein A eluate product stream was captured in a stirred break tank that continuously fed into the low pH viral inactivation (VI) unit operation. Acid titrant was continuously added into the product stream for the entirety of the process. Manual pH control inline titration was performed to meet a pH specification of 3.4–3.6. The subsequent stream was pumped through a static mixer to ensure complete mixing, after which

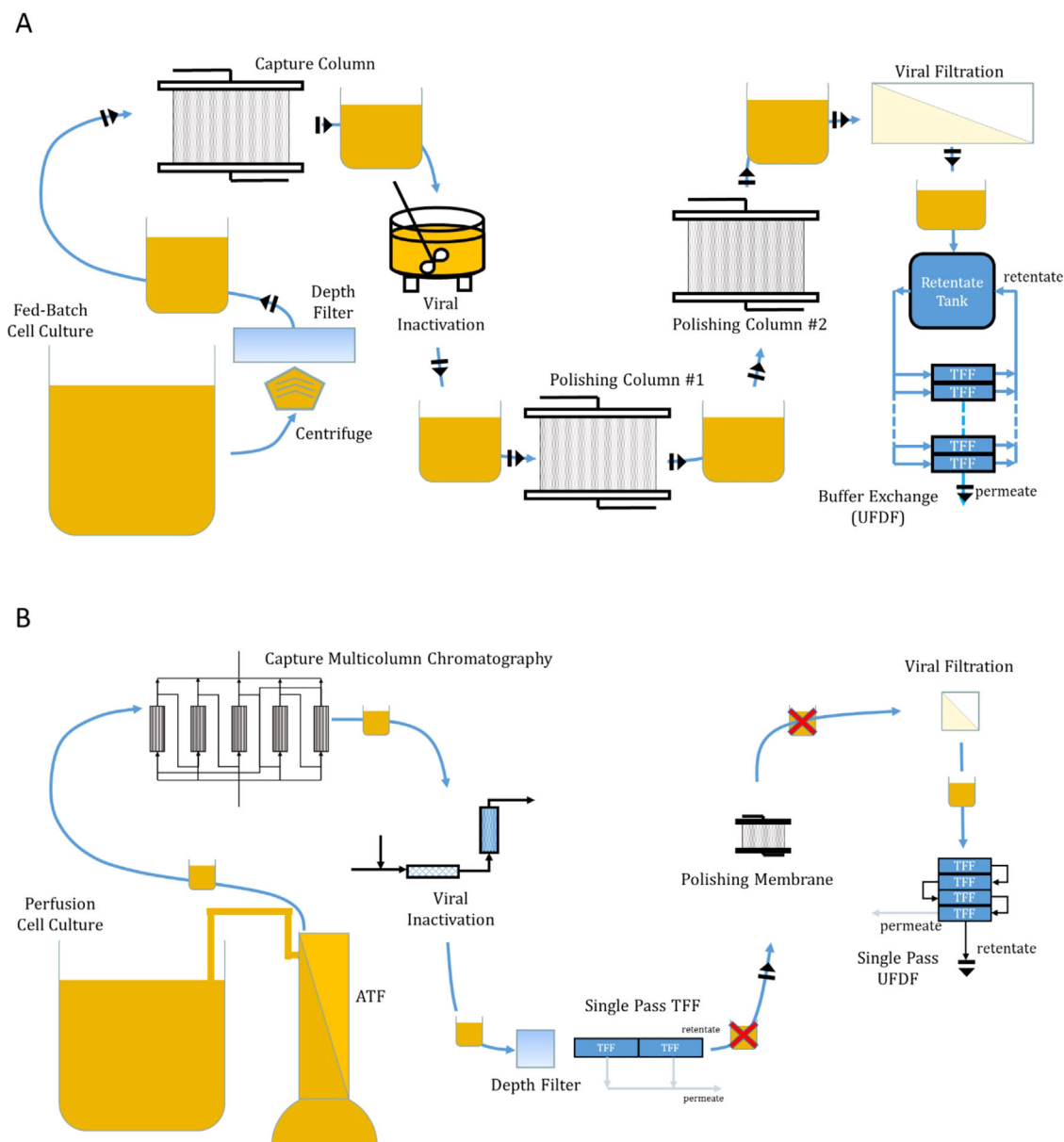


Figure 1. The process flow diagrams of (A) the fed-batch process for drug substance production and (B) experimentally performed fully integrated continuous process. Black bar and arrows (▬) represent the placement of 0.2 µm filters in the process.

the pH was measured inline to confirm and control acid addition. To achieve the target hold time at low pH with continuous flow, the adjusted stream was fed in upflow through a column packed with size exclusion resin. After the required hold time was achieved, the process stream was neutralized to pH 7.4–7.6, with the continuous base flow rate adjusted as needed. Following VI, the product entered a series of filters for the clarification, concentration, anion exchange, and viral filtration steps. The filter train consisted of separate units in series with no line or pressure breaks (Figure 1B). A depth filter was placed first in the filter train, followed by a Millipore SPTFF for product concentration. The Mustang Q membrane provided a low pressure drop option for DNA removal and an orthogonal viral clearance step, and a Planova viral filter was used for

primary viral clearance. The final unit operation incorporated a series of SPTFF membranes to facilitate single pass diafiltration and final concentration. With the current design, only three stages were required to achieve a 99.75% buffer exchange, equivalent to a six diavolume recirculated TFF process. A fourth SPTFF membrane was added to the train to achieve final concentration of the pre-formulated bulk (PFB).

3.2. Design Considerations of the Continuous Process

Several unconventional design considerations were implemented for the fully integrated continuous process. A comparison between Figure 1A and B depict some notable, beneficial

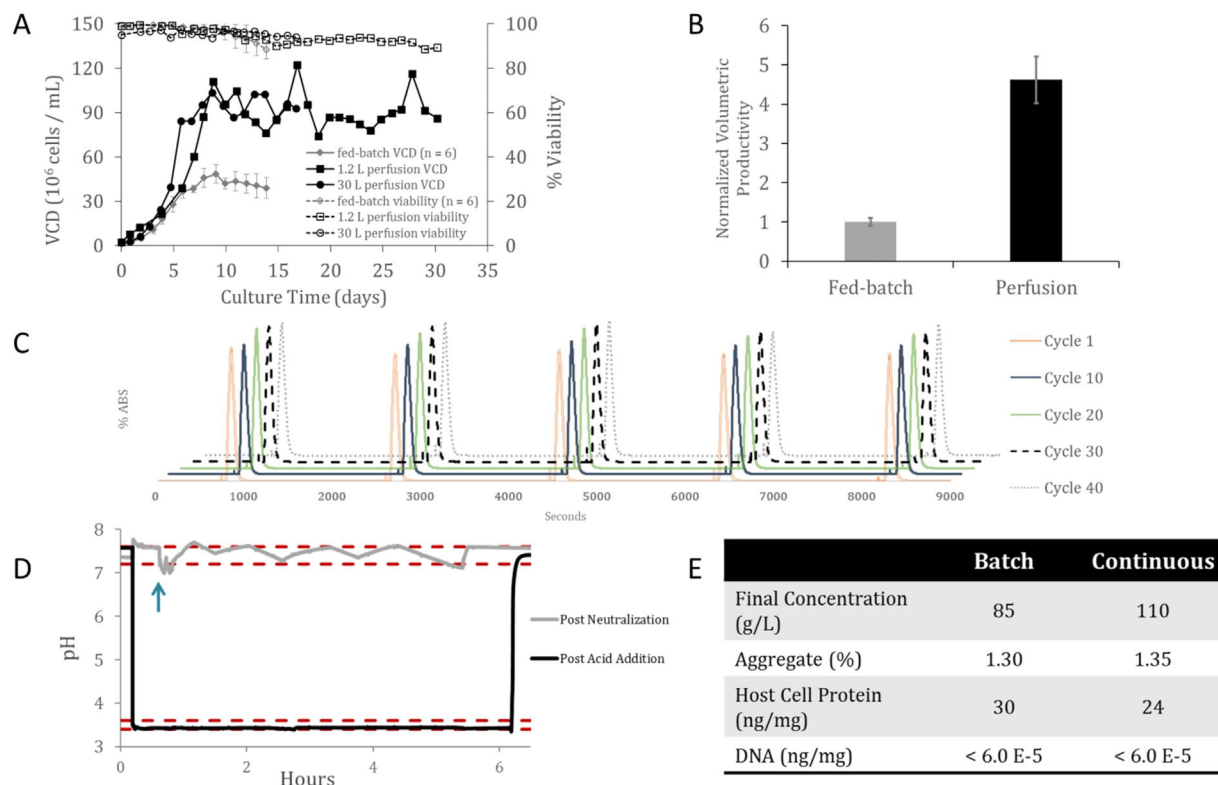


Figure 2. Process performance data comparison of a fed-batch versus continuous process, including (A) viable cell density (VCD) and viability, (B) normalized volumetric productivity, (C) elution peaks from each of the five columns in the multicolumn chromatography unit operation, (D) pH trend of the continuous viral inactivation step of both low pH and neutralized product over the length of the run with the arrow indicating the point of a slight pH excursion after neutralization, and (E) aggregate, host cell protein and DNA in preformulated bulk. The fed-batch VCD and viability plots represent the averages across multiple scales (2–500 L; $n = 6$). Average volumetric productivity for continuous process at the two different scales ($n = 2$) was normalized to the average fed-batch data ($n = 6$), and one way analysis of variance by JMP13 (SAS, Cary, NC, USA) showed $p < 0.01$. All error bars indicate one standard deviation.

differences. Most notable was the ability to operate without dependence on large pool tanks. In the continuous processing approach, drastically lower flow rates, slower processing, and linked unit operations resulted in complete removal or reduction of intermediate product pooling. The break tanks pictured in Figure 1B were only required for pressure relief and typically contained around 100–200 mL at any given point during operation. For the flow through any exchange and viral filtration steps, the break tanks were removed. Furthermore, perfusion production resulted in lower aggregate levels of the product after harvest compared to fed-batch, allowing an additional aggregate removal polishing step to be removed in the continuous process (data not shown). Another difference between batch and continuous schemes was the size reduction of all downstream steps. For example, typical Protein A column requirements to process a 6 days of perfusion product from a 50 L bioreactor would be on the order of 20–30 cm diameter with a 20 cm bed height, or 6–14 L of resin. This column would only be operated for 1 or 2 cycles to process a 6 day collection from the bioreactor. Using multicolumn chromatography, only five 2.2 cm diameter columns at a 5 cm bed height, or 100 mL of resin, was required. These reduced columns are subsequently cycled more rapidly than the larger, batch column design. Instead of one or two cycles per week, the small columns are operated for 10–11

cycles per day, allowing resin lifetime to be reached in about 3 weeks of continuous operation. The case for prepacked, disposable columns becomes attractive when resin lifetime can be achieved within a campaign. Finally, the continuous processing scheme enabled reduction of the number of 0.2 μ m filters from 11 in batch to 2 in continuous. This is due to the lack of intermediate holds in continuous operation.

Benefits more specific to individual unit operation design were also observed. A comparison of batch to perfusion production methods demonstrates improved productivity. To allow for a simple production comparison, the volumetric productivity was calculated: for perfusion this was the total mass produced per bioreactor volume during harvest divided by total run duration; for fed-batch this was the final titer divided by total run duration. This normalization clearly shows that the perfusion volumetric productivity is 4.6-fold greater than that of fed-batch, as seen in Figure 2B.

Using multicolumn chromatography for Protein A capture not only reduced the resin requirement but demonstrated consistency. Elution peak data is shown in Figure 2C. The Protein A elution peaks from each of the five columns are overlaid for cycles 1, 10, 20, 30, and 40 to visually demonstrate the consistency and steady state operation seen in multicolumn mode over the course of multiple days. The elution pool

concentration reached a steady state of 13 g L^{-1} , and yield across the system averaged 96%. Product quality attributes were monitored over time and demonstrated consistency (data not shown). The Protein A product pool volume in the break tank was well-mixed and kept at the equivalent of five elutions to allow the break tank bulk to dampen the pH and concentration variability from subsequent elutions. This allowed a consistent feed to be propagated on to viral inactivation.

Design of the viral inactivation hold column was a new requirement for continuous application. In short, pH adjustment occurred before and after a flow through “hold” design, similar to other theorized methods.^[27] While other flow through viral inactivation approaches were tested, use of a column to achieve required residence time was a compact, inexpensive, scalable, and easily designed solution compared to other technologies or periodic processes found in literature.^[28,29] The column dimensions and flow rate were sized such that the residence time matched the target low pH hold time for viral inactivation. This hold time was dictated by previous viral clearance studies, and is typically on the order of 30–60 min. This column design allows for easy scalability, as the residence time in a column is well understood. While the residence time in the column is not discrete, a transition analysis study was performed to ensure the residence time variability remained within hold time specifications (data not shown). HETP and asymmetry analysis of the packed column were performed before and after processing to ensure the integrity of the column. In addition, the column inlet pressure was monitored over the course of the experiment to help identify potential channeling or fouling, although none were observed. SEC resin was chosen as an inert packing material, as the column purpose was for hold time and not for a purification operation. Additionally, SEC resin was selected as an inexpensive consumable, as it is composed simply of agarose with no additional moieties or functionalities. The pH was measured inline post acidification but before the column, and again in the break tank where neutralization occurred. These measurements both ensured the in-process fluid was in the specified range, as well as fed the feedback control loop controlling the acid and base addition pumps. The trend in Figure 2D shows steady state consistency at the low pH target, whereas more variability was seen during neutralization. This variability is most likely due to pump limitations on flow rate resolution.

The filter train was also adapted to better fit the continuous process. The first filter selected was a depth filter to remove potential particles or turbidity that has been associated with product neutralization after a low pH hold. However, the depth filter may not have been required for this process since VI neutralization turbidity did not rise above 10 NTU. After the depth filter, a SPTFF system was used for process intensification to concentrate the product approximately twofold to both reduce the volume and extend the in-process time of the downstream anion exchange and virus removal filters. In the batch process, the bind and elute CEX operation yielded inherent concentration of the product intermediate, which the SPTFF imitated. Downstream of the SPTFF step was the first $0.2 \mu\text{m}$ filter in the process. The purpose of the $0.2 \mu\text{m}$ filter was to ensure a clean process stream prior to the anion exchange membrane chromatography step and viral filtration. The Mustang Q

membrane was selected instead of a column due to low pressure drop, which was crucial with so many filters inline with no pressure break. The final filter was a Planova viral filter, which was selected due to the more consistent flux profile observed over the filter lifetime compared to a Viresolve Pro solution. As the goal of continuous processing is to achieve steady state, a filter with visible flux decay over the course of a batch will bottleneck the continuity. Oversizing a viral filter could solve this problem as well, but results in a larger hold up volume of the system, so a hollow fiber filter was selected for robustness. A key optimization parameter was the size of each filter: although many sizes were available for “small scale” operation, the hold-up volume, lifetime, and pressure/flow properties were considered. A smaller filter equates to a smaller hold-up volume, which was advantageous for quick start up and shut down, as well as reduced overall residence time within the purification train. However, since smaller filters must be replaced frequently, it was less disruptive to implement larger filters. Larger filters were also oversized for the process flowrates, which resulted in a low pressure drop and enabled multiple filters to be connected in series.

The engineering, design of experiments, and optimization of the novel MedImmune single pass diafiltration (SPDF) design is covered in Pezzini et al.^[26] The product was concentrated >7.5 -fold across a SPTFF stage and removed $>86\%$ of the original feed buffer. The high concentration retentate was diluted in a break tank with diafiltration buffer to the original feed concentration. This concentration and dilution constituted one stage, and buffer exchange requirements were met by placing multiple stages in series.

Final PFB concentration was 110 g L^{-1} . Hourly samples were taken from the PFB to demonstrate steady-state operation and consistency. PFB product quality attributes are summarized and compared to batch in Figure 2E. The final concentration target was designed to be higher in the continuous case, which was to demonstrate the capability and flexibility of the SPTFF. The aggregate content of the continuously processed PFB was nearly identical to that of the batch process, even with the removal of the secondary polishing step. Host cell protein content and DNA levels were also comparable. Cumulatively, these data demonstrated the potential of a fully integrated continuous process.

3.3. Production Throughput and COGm Analysis

In the pilot scale demonstration outlined in the previous section, it was clearly shown that fully integrated continuous processing delivered comparable product quality to the batch process. Also, this continuous process generated more productive unit operations, resulting in reduced equipment size. **Table 1** captures the details associated with sizing and equipment capabilities. As seen in the “Unit Operation Throughput” column of the experimental continuous run, the first three unit operations were undersized compared to the last five, due to equipment availability constraints. The multicolumn system flow rate capabilities were undersized relative to the mass flow rates of the other operations. However, this mismatch was managed by delaying the initiation of subsequent unit operations that followed the Protein A capture. The current

Table 1. Capabilities of the pilot scale fully integrated continuous process based on consumable sizing.

Equipment	Experimental		Ideal	
	Equipment size	Unit operation throughput (g h^{-1})	Equipment size	Unit operation throughput
Bioreactor	30 L	2.6	200 L	12 g h^{-1} (1 kg every 4 days)
ATF	6 m^2		10 m^2	
Protein A columns	2.2 cm diameter 5 cm bed height	1.7	4.4 cm diameter 5 cm bed height	
Viral inactivation column	5 cm diameter 20 cm bed height	12	5 cm diameter 20 cm bed height	
Single pass TFF	$3 \times 88 \text{ cm}^2$		$3 \times 88 \text{ cm}^2$	
Anion exchange membrane	10 mL		10 mL	
Viral filter	0.12 m^2		0.12 m^2	
Single pass UDF	$4 \times 0.065 \text{ m}^2$		$4 \times 0.065 \text{ m}^2$	

The ideal scenario shows adjusted sizing of the rate limiting steps to match the throughput across the entire process train.

SPDF design already used the smallest membranes available, so further scale down was not possible. Therefore, the appropriate mass flowrate to fully utilizing the existing SPDF system was 12 g h^{-1} . A theoretical calculation was done to determine the scales of the bioreactor, ATF, and MCC system that match this mass flowrate of the downstream process train ("Ideal" columns in Table 1). This calculation predicts that the bench scale system has the capability of processing a 200 L perfusion bioreactor at a steady-state mass flow rate of 1 kg of mAb in 4 days.

Another benefit of continuous processing is the additional dimension of scaling and throughput flexibility. To scale up in the batch processing scenario, all the equipment and consumables are scaled up to match the desired batch size. Should an unexpected loss in yield or an increase in demand occur, an additional batch must be run to meet the production requirements. In the case of continuous processing, the process duration is extended until the required demand is met. Variable process duration allows scaling by time in addition to scaling by volume. Additionally, smaller purification equipment facilitates the implementation of single use components, which reduces the need for CIP and SIP systems. These scenarios and their impact on cost of goods per gram of product produced are explored in the next section.

3.4. Future Facility Modeling

The continuous process successfully demonstrated at pilot scale was extrapolated to commercial scale and compared to the fed-batch production method. The processes were modeled in BioSolve in order to translate improved productivity and single use implementation into tangible monetary savings in a commercial, greenfield scenario. BioSolve is an Excel-based software with an extensive built-in cost data library of equipment, consumables, materials, and capital components that enables the user to model cost of goods calibrated to the throughput production capability. The normalized cost of goods manufactured (COGm) per gram output allowed for easy comparison of various manufacturing scenarios.

Detailed process inputs were based on the PFDs illustrated in Figure 1. In the baseline batch scenario, the plant was designed to contain four 12 500 L bioreactors operated in a staggered

manner and fed serially into one purification train. An average titer of 5.6 g L^{-1} was achieved in the fed-batch runs reported in Figure 2A. All equipment was stainless steel, and an 85% facility utilization was assumed. Labor cost was assumed to be dependent only on the number of runs and duration of each run for both continuous and batch processing scenarios, but independent of continuous or batch scenario and process scale. The consumables, such as filters and columns were sized for the maximum process duration of 30 days regardless of the process duration in the continuous processing scenarios.

To compare batch process output to a continuous process, the experimental outputs from the process (performed according to PFD in Figure 1B) were entered into the modeling software, using a 2000 L single use perfusion bioreactor and appropriately sized downstream equipment. The average daily titer of the perfusion bioreactor was experimentally determined to be 1.4 g L^{-1} , and the total culture duration was set at 30 days, which included a 6 day growth phase. The medium usage and labor for the growth phase was included in the cost calculation. All other facility assumptions, such as utilization rate, were kept the same. The facility throughput was fixed to match the batch scenario. For additional insight into facility throughput flexibility, the culture duration of the continuous manufacturing scenario was varied. All of these scenarios are outlined in Table 2.

The COGm output and related cost breakdown for each of these scenarios are illustrated in Figure 3. Based on these calculations, five 2000 L bioreactors were required to yield approximately the same product mass as the batch scenario (shown as normalized throughput in Table 2). When these two scenarios were compared, the COGm of the continuous scenario was 15% lower than that of the batch scenario, as shown in Figure 3. Notably, the capital requirement was reduced by over 50% (Figure 3) when compared to the 30 day culture duration, which directly translates into reduced facility build time. While some increase in materials cost was seen due to the large volume of cell culture medium required for perfusion, a similar reduction was observed for all other categories with the continuous manufacturing scenario.

Two additional continuous scenarios provided additional insight into facility flexibility by relating culture duration of the perfusion production method to throughput and resulting COGm. In the first variation, culture duration was shortened to

Table 2. Plant throughput analysis matching batch production scale.

	Number of bioreactors	Bioreactors size (kL)	Culture duration (days)	Batches per year	Normalized throughput
Batch	4	12.5	14	60	1.00
Continuous	5	2	17	85	0.79
	5	2	30	50	1.02
	5	2	42	35	1.07

The top three scenarios depict a fixed throughput with variable reactor number and run duration, while the bottom three scenarios depict a fixed facility design and variable run durations and its impact on plant throughput.

17 days, including a 6 day start up period, aligning with the 30 L-scale continuous processing study described above. The number of batches per year was subsequently increased; however, the throughput decreased by 20% due to the cycle time between runs. When the culture duration was instead extended to 42 days, throughput was marginally increased, with a proportional decrease in COGm. Increasing culture duration beyond 42 days yielded diminishing returns but remained advantageous compared to the batch scenario. This model was specific to the process derived for the experimentally tested monoclonal antibody, but the conclusions were in line with other cost modeling exercises.^[17–19,30,31]

4. Discussion

In the current study, the concept of a fully integrated, continuous pilot scale operation was successfully demonstrated. The perfusion bioreactor operated for 17 days, instead of the intended 30 days, and was purified by a fully integrated downstream train consisting of all unit operations from primary capture through UFDF. However, cell culture performance up to this time performed within expected ranges of bench-scale experiments (Figure 2A) and would have achieved a target of 30 days otherwise. Despite this failure, the continuous downstream purification train was able to continue operation and propagation of recovered material beyond the time of ATF

failure. In a batch cultivation, such an adverse event would have resulted in the loss of the entire batch, whereas in continuous processing, material could be recovered from the successful days of operation (17 days in this instance). The same benefits apply to a downstream unit operation, where failure at any given step does not compromise the entire batch, but instead can be isolated.

Overall, it can be seen from this proof of concept study that a fully integrated continuous process would be attractive both in terms of reduced capital cost for facility build, and cost of manufacture, as seen in Figure 3, and comparable purity, as seen in Figure 2E. An overall COGm reduction of 15% can be realized by switching from batch to continuous production. Some caveats and limitations were imposed by BioSolve; in an ideal scenario, a facility containing five production bioreactors would involve staggering each bioreactor across the production period so that once the continuous steady state harvest begins there would be up to four bioreactors generating harvest material and one bioreactor in cell growth phase or between shutdown and startup. Only a fixed run duration could be entered into the model for each scenario. Therefore, the model also included the assumption that all runs reached the target run duration defined in each scenario.

The migration to single use technologies, enabled by continuous manufacturing, promotes faster facility build and a more flexible facility design. Elimination of CIP systems minimizes failure points associated with CIP operation, maintenance, and installation. It has also been conceptually proposed to design future facilities in an open ballroom layout, instead of the separate suite for each set of unit operations.^[32] This design not only minimizes construction time, but also increases flexibility in manufacturing throughput and the ability to easily adapt equipment for multi-product scenarios with reduced capital investment. This results in further COGm reduction in addition to increased volumetric productivity. The open ballroom concept eliminates the majority of facility fit constraints found in dedicated product facilities, such as buffer tank restrictions or insufficient bioreactor capacity. It also allows for quick implementation of additional equipment without lengthy shut down procedures.

In addition to the above quantitative benefits, continuous processing can bring other benefits that ease the transition from clinical to commercial manufacturing. In batch processing, a commercial manufacturing campaign would require a much larger vessel than a Phase I clinical campaign. In continuous processing the increase in demand from Phase I to commercial production can be met by extending the run duration (i.e., scale

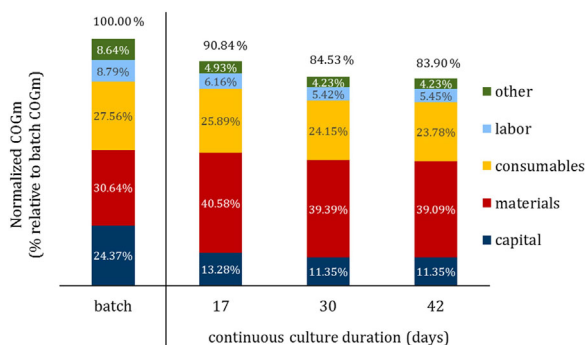


Figure 3. Cost of goods analysis for batch and continuous manufacturing scenarios. The batch scenario represents 12 500 L production bioreactors with an appropriately sized purification train to process all bioreactors: the continuous scenarios represent 2000-L production bioreactors with various culture durations and processed by a single appropriately sized downstream train. Percent COGm for each category is normalized to the batch scenario.

out by time). The efforts required for facility fit and scalability assessment, as well as scale-down model development, can be drastically reduced by eliminating the need to further scale up beyond Phase I clinical manufacture.

Generalization of COGm reduction due to a switch to continuous manufacturing is not possible for all drug products, as indicated by the higher COGm in the continuous scenario at the commercial stage.^[17,25] Relative COGm between the two scenarios is a function of assumptions for each model, including relative volumetric productivity, bioreactor scale, number of bioreactors, purification throughput, and run duration. The work presented by Pollock et al.,^[17] made an assumption for the annual number of batches of $20 \times 20\,000$ L for the fed-batch scenario and 10×1500 L for the continuous scenario. The model presented in this work is based on a batch facility with $4 \times 12\,500$ L bioreactors that operates a total of 60 bioreactor runs per year and a fully integrated continuous facility with 5×2000 L bioreactors with a run duration of 30 days and a total of 50 bioreactor runs per year. Therefore, the ratio of the batch and continuous bioreactor process volumes is lower in the model presented here, which diminishes the advantage of the batch scenario in the economy of scale. One other key assumption in this study is the 5 to 1 ratio for the number of bioreactors to the number of purification trains, with the flexibility to pool up to five bioreactors into a single purification train, based on the purification throughput demonstrated at the pilot scale. Both models, however, agree that the cost of cell culture materials is significantly higher for the continuous scenario, primarily due to an increase in medium usage per unit mass of drug substance produced in the perfusion process. This increased cost is offset by the sizable decrease in downstream materials, particularly when applied to the most expensive material, Protein A resin. This benefit is capitalized on in hybrid approaches, as noted by others.^[17,19,33] There is also a discrepancy in the relative labor cost per gram of drug substance: the scenario presented by Pollock et al. reports increased labor cost per gram of product in the continuous scenario in contrast to the lower value presented here. The main driver for the reduction in labor cost per gram of product in the continuous scenario in this work is the 4.6-fold increase in productivity by switching from batch to continuous manufacturing.

Development and optimization of a continuous process will require focus on different aspects as compared to that of a batch process. As the COGm model indicates, contribution of materials cost increases for a continuous process. This result is expected as the medium usage per unit mass of product is much greater. In addition to increasing volumetric productivity, upstream development should also place more focus on reducing the medium usage by minimizing the perfusion rate. Another benefit of a lower perfusion rate would be the ability to scale up to larger vessels; the maximum ATF filter area currently available limits the harvest rate due to the maximum allowable membrane flux. It is also important to understand the characteristics of the clones that perform well in a high cell density perfusion environment, although such information is yet to be available. It will be worth evaluating whether the existing high throughput clone selection methods

in batch cultures are suitable for continuous perfusion cultures. Downstream process development will remain largely unchanged, with process parameters still being defined by small batch experiments. Conversion to process and production scale will require less of a size change compared to batch, but will require transitional effort to become continuous. In the long run, though, overcoming these transitional risks will reap the rewards of improved productivity, improved COGm, and other benefits associated with continuous processing.

Downstream runs of extended durations used for continuous manufacturing will require the purification process to maintain bioburden and endotoxin control. Disposable, gamma irradiated systems in combination with solution sanitization procedures will be required. The ability to quickly identify any sign of microbial growth with rapid at-line microbial and endotoxin detection systems can enable continued downstream production with re-sanitization of the system and/or replacement of disposable items. It is recognized that additional bioburden and endotoxin testing would be required for a continuous process when compared to batch, which may incur additional costs not captured in this analysis.

As illustrated in this work, continuous processing offers many opportunities for the biopharmaceutical industry to reduce manufacturing costs and increase throughput. Continuous processes require shifting the focus for process development and will require a change in some operating requirements. However, all the future challenges can be potentially overcome as demonstrated by the continued advancement of novel continuous process technologies reported in recent years,^[34–36] and continuous manufacturing will likely become common practice in the coming future.

Abbreviations

APF, animal protein free; ATF, Alternating tangential flow; CHO, Chinese hamster ovary; CIP, clean in place; CMC, chemistry, manufacturing, and controls; COGm, cost of goods manufactured; CSPR, cell specific perfusion rate; HPLC, high performance liquid chromatography; HPSEC, high performance size exclusion chromatography; LLOQ, lower limit of quantitation; mAb, monoclonal antibody; MCC, Multi column chromatography; PES, polyether sulfone; PFB, preformulated bulk; PFD, process flow diagram; SIP, steam in place; SPTFF, single pass tangential flow filtration; SPDF, single pass diafiltration; SUB, single use bioreactor; TFF, tangential flow filtration; VCD, viable cell density; VI, viral inactivation; VVD, volume perfused per reactor volume per day.

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Conflict of Interest

The authors declare no financial or commercial conflict of interest.

Keywords

COGm, future facility, integrated continuous bioprocessing, multicolumn chromatography, perfusion

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- [1] V. Anticetti, *BioProcess Int.* **2009**, 7, 4.
- [2] H. L. Levin, J. E. Lilja, R. Stock, H. Hummel, S. D. Jones, *BioProcess Int.* **2012**, 10, 20.
- [3] U. Gottschalk, K. Brorson, A. A. Shukla, *Nat. Biotechnol.* **2012**, 30, 489.
- [4] N. G. Anderson, *Org. Pro. Res. Dev.* **2001**, 5, 613.
- [5] N. Fletcher, *Manuf. Chem.* **2001**, 24, 26.
- [6] H. Thomas, *Chem. Eng.* **2008**, 805, 38.
- [7] C. K. S. Ng, F. Rousset, E. Valery, D. G. Bracewell, E. Sorensen, *Food and Bioproducts Processing*. **2014**, 92, 233.
- [8] V. Warikoo, R. Godawat, K. Brower, S. Jain, D. Cummings, E. Simons, T. Johnson, J. Walther, M. Yu, B. Wright, J. McLarty, K. P. Karey, C. Hwang, W. Zhou, F. Riske, K. Konstantinov, *Biotechnol. Bioeng.* **2012**, 109, 3018.
- [9] S. G. Desai, *J. Biotechnol.* **2015**, 213, 20.
- [10] B. G. Boedeker, *Semin Thromb Hemost.* **2001**, 27, 385.
- [11] B. Kelley, M. Jankowski, J. Booth, *Hemophilia*. **2010**, 16, 717.
- [12] X. Gjoka, R. Gantier, M. Schofield, *BioPharm. Int.* **2017**, 30, 32.
- [13] V. Girard, N. Hilbold, C. K. S. Ng, L. Pegon, W. Chahim, F. Rousset, V. Monchois, *J. Biotechnol.* **2015**, 213, 65.
- [14] A. K. Dutta, T. Tran, B. Napadensky, A. Teella, G. Brookhart, P. A. Ropp, A. W. Zhang, A. D. Tustian, A. L. Zydne, O. Shinkazh, *J. Biotechnol.* **2015**, 213, 54.
- [15] P. Gronemeyer, R. Ditz, J. Strube, *Bioeng.* **2014**, 1, 188.
- [16] A. Patel, BioSC in a fully integrated continuous Mab manufacturing process. *BPI Industry Yearbook. 2017-2018.* **2017**, 62.
- [17] P. Pollock, J. Coffman, S. V. Ho, S. S. Farid, *Biotechnol. Progr.* **2017**, 33, 854.
- [18] S. Xu, J. Gavin, R. Jiang, H. Chen, *Biotechnol. Progr.* **2017**, 33, 867.
- [19] A. Xenopoulos, *J. Biotechnol.* **2015**, 213, 42.
- [20] K. B. Konstantinov, C. L. Cooney, *J. Pharm. Sci.* **2015**, 104, 813.
- [21] A. L. Zydne, *Curr. Opin. Chem. Eng.* **2015**, 10, 8.
- [22] R. Hernandez, *BioPharm Int* **2015**, 28, 20.
- [23] M. Brower, D. Pollard, F. Hung, Protein Refinery Operations Lab (PRO Lab): A sandbox for continuous protein production & advanced process control. Integrated Continuous Biomanufacturing II conference. **2015**. Oral presentation.
- [24] S. Klutz, J. Magnus, M. Lobedann, P. Schwan, B. Maiser, J. Niklas, M. Temming, G. Schembecker, *J. Biotechnol.* **2015**, 213, 120.
- [25] S. Farid, B. Thompson, A. Davidson, *mAbs* **2014**, 6, 1357.
- [26] J. Rucker-Pezzini, L. Arnold, K. Hill-Byrne, T. Sharp, M. Avazhanskiy, C. Forespring, Single pass diafiltration integrated into a fully continuous mAb purification process. *Biotechnol. Bioeng.* **2018**, Article in Press. <https://doi.org/10.1002/bit.26608>
- [27] S. Johnson, M. Brown, S. Lute, K. Brorson, *Biotechnol Bioeng.* **2017**, 114, 21.
- [28] S. Klutz, M. Lobedann, C. Bramsiepe, G. Schembecker, *Chem. Eng. Process.: Process. Intens.* **2016**, 102, 88.
- [29] R. Orozco, S. Godfrey, J. Coffman, L. Amari kwa, S. Parker, L. Hernandez, C. Wachuku, B. Mai, B. Song, S. Hoskatti, J. Asong, *Biotechnol. Prog.* **2017**, 33, 954.
- [30] J. Walther, R. Godawat, C. Hwang, Y. Abe, A. Sinclair, K. Konstantinov, *J. Biotechnol.* **2015**, 213, 3.
- [31] J. Hummel, M. Pagkaliwangan, X. Gjoka, T. Davidovits, R. Stock, T. Ransohoff, R. Gantier, M. Schofield, *Biotechnol. J.* **2018**, Article in Press. <https://doi.org/10.1002/biot.201700665>
- [32] M. Pelletier, Impact of closed and continuous processing on biopharmaceutical facility layouts, Integrated Continuous Biomanufacturing II conference. **2015**. Oral presentation.
- [33] P. Bunnak, R. Allmendinger, S. V. Ramasamy, P. Lettieri, N. J. Titchener-Hooker, *Biotechnol. Prog.* **2016**, 32, 1324.
- [34] F. Steinebach, T. Müller-Späh, M. Morbidelli, *Biotechnol. J.* **2016**, 11, 1126.
- [35] N. Walch, A. Jungbauer, *Biotechnol. J.* **2017**, 12, 1700082.
- [36] N. Kateja, H. Agarwal, A. Saraswat, M. Bhat, A. S. Rathore, *Biotechnol. J.* **2016**, 11, 1320.